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TECHNICAL NOTES

Characterization of polymorphic microsatellite loci for the Tawny Pipit *Anthus campestris* and their utility in other steppe birds

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Abstract New microsatellite loci for Tawny Pipit *Anthus campestris* were isolated from a genomic library. We were able to unambiguously score six loci: two were dinucleotide, one trinucleotide, two tetranucleotide and one pentanucleotide that turned out to be sex-linked. Four out of six loci were polymorphic with 7–23 alleles in our population and an observed heterozygosity ranging between 0.286 and 0.936. Cross-utility of these markers was tested in other 17 steppe-bird species of six families. In addition, 16 microsatellite loci developed for other species were tested for cross-species amplification in *A. campestris*. Eight microsatellite markers were successfully amplified; seven of them were polymorphic with 2–43 alleles and an observed heterozygosity of 0.040–0.863. Overall, 14 functional locus markers have been characterized for *A. campestris* that could be useful for future studies of paternity, genetic variability and population structure.

Keywords *Anthus campestris* · Microsatellite · Cross-amplification · Motacillidae · Steppe birds · Tawny Pipit

Introduction

The genus *Anthus* contains 43 passerine species, although specific microsatellite markers are only available for Sprague's Pipit *Anthus spragueii* (Crawford et al. 2009), and cross-species amplification has only been tested for Berthelot's Pipit *Anthus berthelotii* (Illera et al. 2007). The Tawny Pipit *Anthus campestris* has a wide breeding distribution range from Western Europe and North Africa to Mongolia (Alström and Mild 2003). European populations have suffered a sharp decline during the last decades (Birdlife International 2010), a common trend among steppe and farmland birds derived from land management changes (Pain and Pienkowski 1997; Donald et al. 2001; Benton 2007). As a consequence of this decline, *A. campestris* populations have become increasingly isolated, and populations on average are much smaller than those recorded historically (Koleček et al. 2010). The scientific information available for this species is noticeably low, even though the species has become recently extinct from some European countries (Van Turnhout 2005). At this point, identifying current diversity and connectivity among populations of *A. campestris* is one goal for species conservation. Microsatellites are neutral nuclear genetic markers that can be utilized to identify the genetic diversity of a given population (Bruford and Wayne 1993), as well as relatedness among populations (Jarne and Lagoda 1996), and could be of great utility for the study and conservation of populations of this grassland bird (DeSalle and Amato 2004).

Our aims were to develop polymorphic microsatellites for *A. campestris* and to test its utility for other steppe-bird species with none or few available microsatellite markers. Additionally, in order to gather more informative markers, we tested non-specific microsatellite cross-amplification in *A. campestris*.

Methods

Microsatellite loci were isolated from a genomic library enriched for $(GATA)_7$, $(AAAG)_7$, $(AC)_{13}$, $(AGC)_8$ and

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Table 1 Microsatellite loci isolated from *A. campestris* and cross-species microsatellite loci amplified in *A. campestris*

Species isolate reference	Locus	Primer sequence (5′–3′)	Accession no.	Repeat motif	T^a (°C)	[$MgCl_2$] (mM)	Size (bp)	N_A/Ns	H_o	H_e	F_{IS}
Anthus campestris	Aca2	F:AAGGAGGGTGGGTTCAGAGT R:6FAM-ACCCACAAGCTGAACTCTGT	HM136874	$(TGC)_8$	58	2	103–157	14/51	0.823	0.827	0.004
	Aca9	F:TGGACCTCTTGAGAGCAGTG R:6FAM-CTGCATTTCTACCCGTGATG	HM136878	$(ATCC)_{15}$	49	2	199–257	23/47	0.936	0.938	0.002
	Aca8	F:TATCCCACATCTCTGTGAGC R:VIC-AGAGGAGAGGAGCAGAGTCC	HM136877	$(CTATT)_9$	58	2	190–220	7/29 males	0.862	0.807	0.050
	Aca21	F:GCTCTCCAGGTGATGGTCCT R:PET-TCCATGCCCTTGTCCACAGT	HM136879	$(GGAT)_{25}$	49	2	121–219	15/51	0.286	0.558	0.496
	Aca3	F:GTTAAAAATCTATGCATCCAGC R:NED-ACTGAAAAAGCTGCAGACACAC	HM136875	$(CA)_{19}$	46	1.5	155	1/51	–	–	–
	Aca5	F:GCAAATCTGCGTGTGTGA R:NED-TGCTCATACACTACCTGAGCAGA	HM136876	$(GT)_{47}$	40	2	170	1/51	–	–	–
Ficedula hipoleuca (Primmer et al. 1996)	FhU2	F:GTGTTCTTAAAACATGCCTGGAGG R:VIC-GCACAGGTAAATATTTGCTGGGCC	X84361	$(CT)_{12}$	55	1.5	127–141	7/51	0.686	0.603	−0.139
Acrocephalus arundinaceus (Hansson et al. 2000)	Aar8	F:TAGTGATGCCCTGCTAGGTA R:NED-AAGTGCTCCTTAATATTTGGCA	AF234991	$(CA)_{17}(GA)_7$	55	1.5	87	1/47	–	–	–
Malurus cyaneus (Double et al. 1997)	McyU4	F:ATAAGATGACTAAGGTCTCTGGTG R:PET-TAGCAATTGTCTATCATGGTTTG	U82388	$(GT)_{26}$ AT $(GT)_3$	55	1.5	134–167	13/50	0.780	0.891	0.125
Pica pica (Martinez et al. 1999)	Ppi2	F:CACAGACCATTCGAAGCAGA R:NED-GCTCCGATGGTGAATGAAGT	AJ272375	$(GTT)_1(GT)(TT)_1(GT)_9$	50	1.5	234–260	13/51	0.823	0.886	0.072
Parus domesticus (Griffith et al. 1999)	Pdoμ5	F:GATGTTGCAGTGACCTCTCTTG R.VIC-GCTGTGTTAATGCTATGAAAATGG	Y15126	$(TG)_{21}$	50	1.5	200–236	11/51	0.579	0.598	0.083
Catharus ustulatus (Gibbs et al. 1999)	Cu10	F:AAAATGAGGAGAATACTAGGCA R:6FAM-ACTTATTTCAGTCCTAAATTCACC	AF122893	$(CA)_{10}$	46	2.5	138–199	43/51	0.863	0.868	0.006
Acrocephalus sechellensis (Richardson et al. 2000)	Ase19	F:TAGGGTCCCAGGGAGGAAG R:NED-TCTGCCCATTAGGGAAAAGTC	AJ276376	$(CA)_4GA(CA)_5$	55	2	157–167	2/50	0.040	0.039	0.010
Poecile atricapillus (Otter et al. 1998)	PAT MP 2–43	F:ACAGGTAGTCAGAAATGGAAAG R:NED-GTATCCAGAGTCTTTGCTGATG	–	Not published	53	2.5	120–128	4/50	0.260	0.294	0.116

Primer sequence, GenBank accession number (accession no.), repeat motif of cloned allele, annealing temperature (T^a), salt concentration ([$MgCl_2$]), allele size range (size), number of alleles (N_A) for the number of samples tested (N_S), observed heterozygosity (H_o), expected heterozygosity (H_e), fixation index (F_{IS}). Cross- species microsatellite loci tested with unsuccessful amplification for *A. campestris*: Lox8 (Y16825; Piertney et al. 1998), HrU5 and HrU7 (X84090 and X84092; Primmer et al. 1996), Pca4 and Pca7 (AJ279806 and AJ279809; Dawson et al. 2000), PmaC25 (AY260526; Saladin et al. 2003), Phtr2 (EF621527; Fridolfsson et al. 1997) and Cu04 (AF122891; Gibbs et al. 1999)

$(GAT)_8$ repeats, constructed following a procedure modified from Hamilton et al. (1999). Briefly, 50 μg of DNA was digested with 150 U each of *Alu*I, *Bsp*LI, *Rsa*I, *Hinc*II, *Bsh*1236I and *Bsu*RI (Fermentas). Restriction fragments of 500–3,000 base pairs (bp) were excised from an agarose gel and extracted with QIAquick Gel Extraction Kit (Qiagen). Fragments were dephosphorylated with calf intestinal alkaline phosphatase (Fermentas) and ligated to SNX linkers (Hamilton et al. 1999). Purified DNA was then hybridized with biotinylated microsatellite oligonucleotides, and the DNA fragments with repeat sequences were captured with streptavidin-coated magnetic beads (Dynabeads M-280 Streptavidin, Invitrogen) and subsequently amplified by PCR with SNX-F primer. Amplified DNA was ligated into the *Xba*I site of pUC19 (Fermentas), and plasmid constructs were used to electroporate ElectroTen-Blue electroporation-competent *Escherichia coli* cells. Transformed bacteria were planted on LB agar medium and lifted onto nylon membranes, which were probed with DIG-labelled microsatellite motives used in the enrichment. Screened positives with inserts in the range of 300–700 bp were sequenced with forward M13 primer and BigDye Terminator kit version 3.1 in an ABI PRISM 3130xl sequencer (Applied Biosystems). A total of 92 positive clones were sequenced and edited in BioEdit 7.0.4.1 (Hall 1999); 58 clones contained repeat sequences. Of these, 21 were chosen for primer design using the PRIMER3 software (Rozen and Skaletsky 2000). The remaining sequences were discarded due to the presence of very irregular and/or very long repeat sequences. PCR amplifications were optimized for each locus and consisted of an initial denaturation of 3 min at 94°C followed by 35 cycles of 30 s at 94°C, 30 s at the annealing temperature (Table 1) and 30 s at 72°C, plus a final extension of 10 min at 72°C. All PCRs were performed in a 2720 thermal cycler (Applied Biosystems). The reaction mix (10 μL final volume) contained 1.5–2.5 mM $MgCl_2$ (Table 1), 0.25 U of Taq DNA polymerase (Biotools), 0.5 μM of each primer, 0.125 mM of each dNTP and 25–50 ng of genomic DNA. Reverse primers were tagged with fluorescent labels on the 5′-end (6-FAM, NED, PET or VIC; see Table 1). The amplification products were genotyped in an ABI PRISM 3130xl sequencer (Applied Biosystems). Allele sizes were determined according to GeneScan 500-LIZ Size Standard and scored using GeneMapper version 4.0 (Applied Biosystems).

Polymorphism levels in *A. campestris* were tested in a sample of 47–51 adults captured in 2007 from a breeding population in central Spain (41°05′ N, 2°19′ W; Layna Plateaux, Soria). Cross-species amplification of microsatellites isolated from *A. campestris* was tested in three individuals from 17 species (Table 2) using the same conditions described above. Also, 16 non-specific microsatellite loci were tested for cross-amplification in *A. campestris* (Table 1). Blood samples were obtained by venepuncture of the brachial vein and preserved in SET buffer upon DNA extraction. All individuals were sexed using primers 0057F and 002R (Round et al. 2007).

Data were checked for null alleles, stuttering and allele dropouts using MICRO-CHECKER (Oosterhout et al. 2004). The program GENEPOP 4.0 (Rousset 2008) was

Table 2 Test for cross-utility of microsatellite loci isolated from *A. campestris* in 17 species of six different steppe-bird families

Species	Family	Aca2	Aca9	Aca8	Aca21	Aca3	Aca5
Motacilla alba White Wagtail	Motacillidae	+	–	–	–	–	–
Alauda arvensis Skylark	Alaudidae	+	+	+	+	i	i
Alaemon alaudipes Hoopoe Lark		–	–	+	+	+	i
Calandrella brachydactyla Short-Toed Lark		–	+	–	+	–	i
Calandrella rufescens Lesser Short-Toed Lark		–	+	+	+	i	–
Chersophilus duponti Dupont's Lark		+	+	–	+	–	i
Eremophila bilopha Temminck's Lark		–	+	+	+	+	i
Galerida cristata Crested Lark		–	+	+	+	+	i
Galerida theklae Thekla Lark		+	+	+	+	i	i
Ramphocoris clotbey Thick-Billed Lark		+	+	+	+	–	i
Oenanthe hispanica Black-Eared Wheatear	Muscicapidae	+	–	–	+	+	–
Oenanthe oenanthe Northern Wheatear		–	–	+	+	i	i
Oenanthe deserti Desert Wheatear		+	+	+	+	i	–
Cursorius cursor Cream-Coloured Courser	Glareolidae	–	+	–	–	+	i
Pterocles alchata Pin-tailed Sandgrouse	Pteroclidae	–	–	–	–	–	–
Pterocles orientalis Black-Bellied Sandgrouse		–	–	–	+	+	i
Tetrax tetrax Little Bustard	Otididae	+	–	–	+	i	i

+ Positive amplifications, − negative amplifications, *i* unspecific amplifications used to calculate the number of alleles (N_A), observed (H_o) and expected (H_e) heterozygosity and fixation index (F_{IS}). Also, we tested for deviations from Hardy–Weinberg and linkage equilibrium using the log likelihood ratio statistic and 1,000 dememorizations with 100 batches (1,000 iterations per batch) and a significance level of 0.05, or the value obtained after sequential Bonferroni corrections (Holm 1979).

Results and discussion

Of the 21 microsatellite loci tested, six were unambiguously scored of which four were polymorphic. Two loci consisted of dinucleotide repeats; one was trinucleotide, two tetranucleotide and one pentanucleotide (Table 1). A total of 59 alleles (7–23 alleles per locus) were detected for the four polymorphic loci in 51 *A. campestris* (Table 1). Observed and expected heterozygosities ranged from 0.286 to 0.936 and 0.558 to 0.938, respectively (Table 1). No significant linkage disequilibrium between loci or deviations from Hardy–Weinberg equilibrium was found, except for Aca21 that showed a significant excess of homozygotes (F_{IS}= 0.496, P<0.001). This locus was also identified by MICRO-CHECKER as likely having null alleles. All females (n=22) were homozygous for locus Aca8. Further examination of their offspring (data not shown) and male data separately (Table 1) evidenced that this marker might be linked to sex-chromosome Z. Microsatellite repeat motifs and flanking region, including primer sequences, are available on GenBank (accession no. HM136874-HM136879).

The six novel *A. campestris* microsatellites were tested for cross-amplification in 17 species belonging to six bird families (Table 2). Within the species tested, only three had been previously tested for microsatellite amplification: *Alauda arvensis* (Hutchinson and Griffith 2008), *Oenanthe oenanthe* (Kudernatsch et al. 2009) and *Chersophilus duponti* (Méndez et al. 2010). All species were successfully amplified for at least one of the microsatellites analysed, except for the *Pterocles alchata* (Table 2). The most useful markers were Aca21 and Aca9, which rendered successful amplifications in 14 and 10 species, respectively. In contrast, Aca5 showed non-specific bands for most of the species tested (Table 2). Even though levels of variability of the *A. campestris* markers were not tested in the other species, further research should clarify their suitability for future population genetic and parentage analyses.

Eight out of the 16 non-specific loci tested in *A. campestris* were unambiguously scored, and seven of them were polymorphic in our study population (average N_A= 13.286±13.793; range, 2–43; Table 1). Particularly, loci Ppi2 and Pdoμ5 could be useful for studies in the genus *Anthus*, as they have also been proven to be polymorphic in *A. berthelotii* (Illera et al. 2007). The remnant eight loci tested were not useful due to the amplification of multiple bands or no amplification at all (see Table 1).

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